

First Encounters & Social Needs

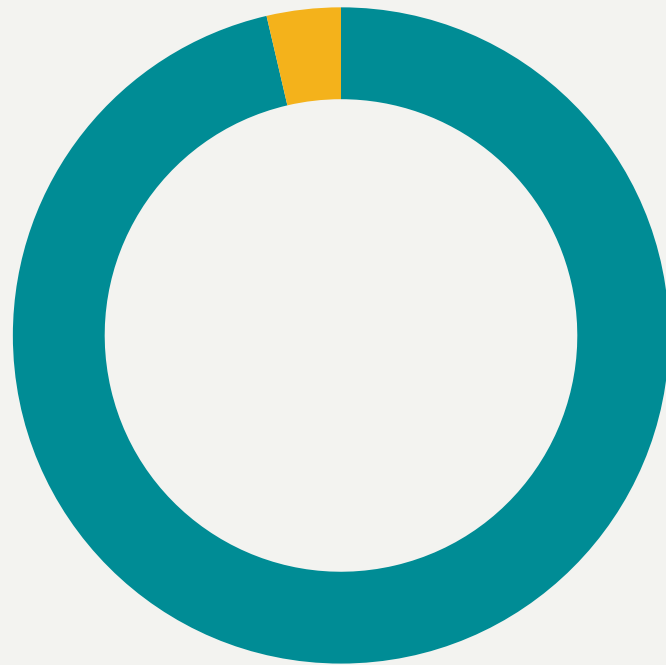
Exploring how first observed chronic disease encounters differ by care setting and recorded social needs.

The8282s / DataFest 2026

2022–2025 records

ED and non-ED entry points

One first observed chronic disease encounter per patient



■ Non-ED ■ ED

84,850

patients in the full cohort

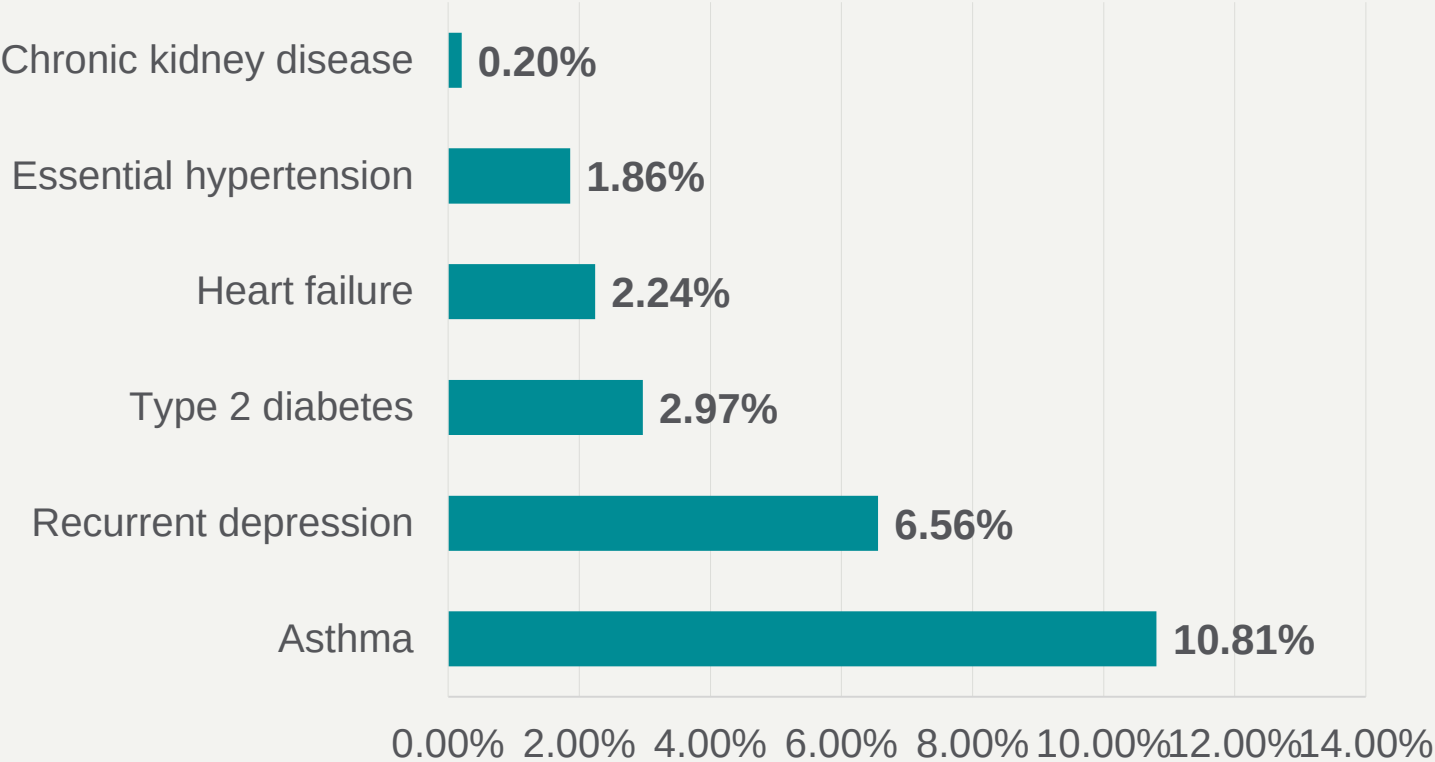
3.64%

3,092 ED first encounters

First observed in this dataset, not necessarily the first diagnosis. Non-ED can include inpatient care.

ED rates by disease

Each patient contributes one first observed encounter per disease

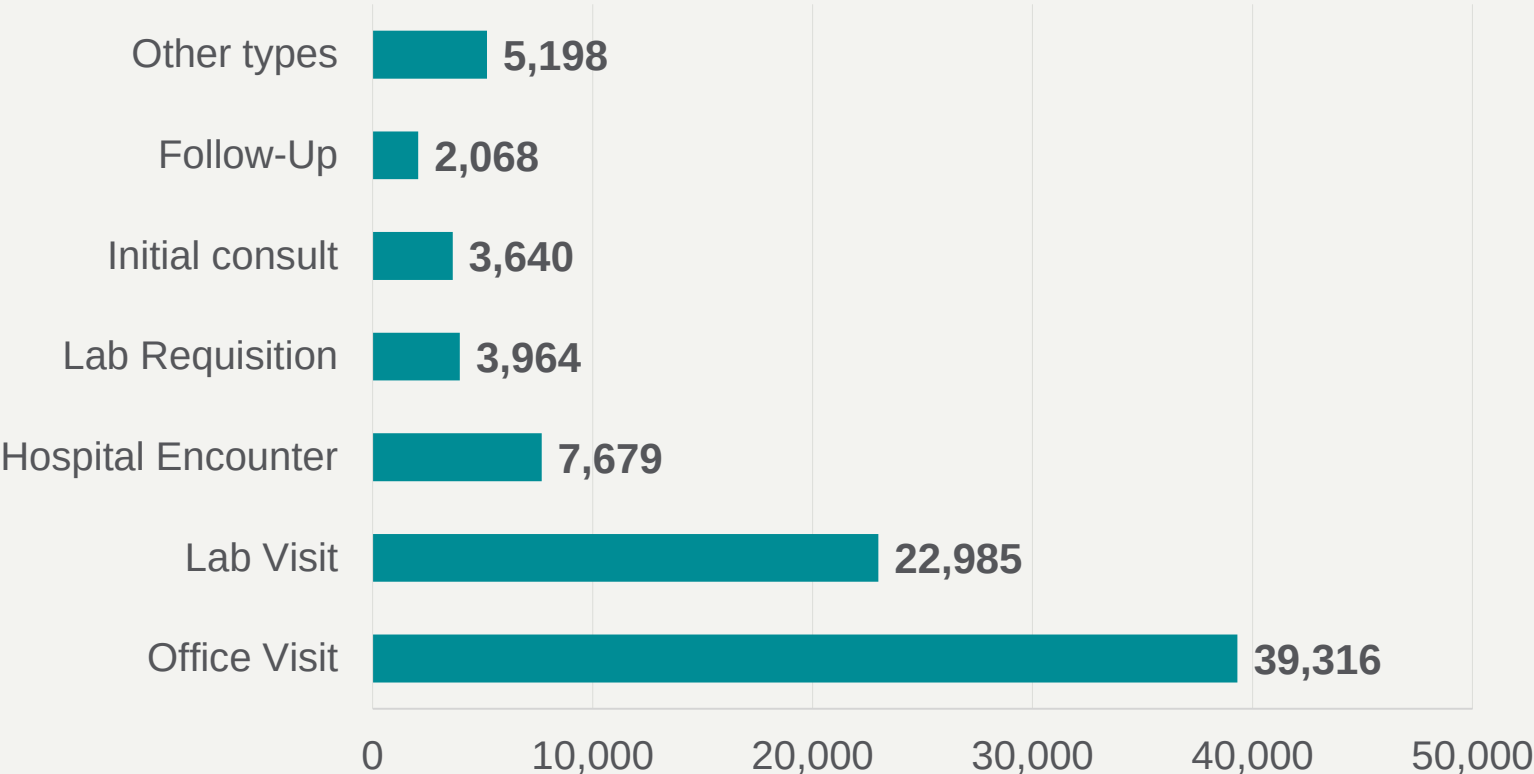


Disease	ED / patients
N18	25 / 12,205
I10	996 / 53,567
I50	162 / 7,228
E11	762 / 25,668
F33	600 / 9,147
J45	1,057 / 9,782

117,597 patient–disease pairs across 84,850 patients. A patient may appear in several disease groups.

First-encounter types

Encounter type describes the record, while the ED flag identifies emergency visits



3,092

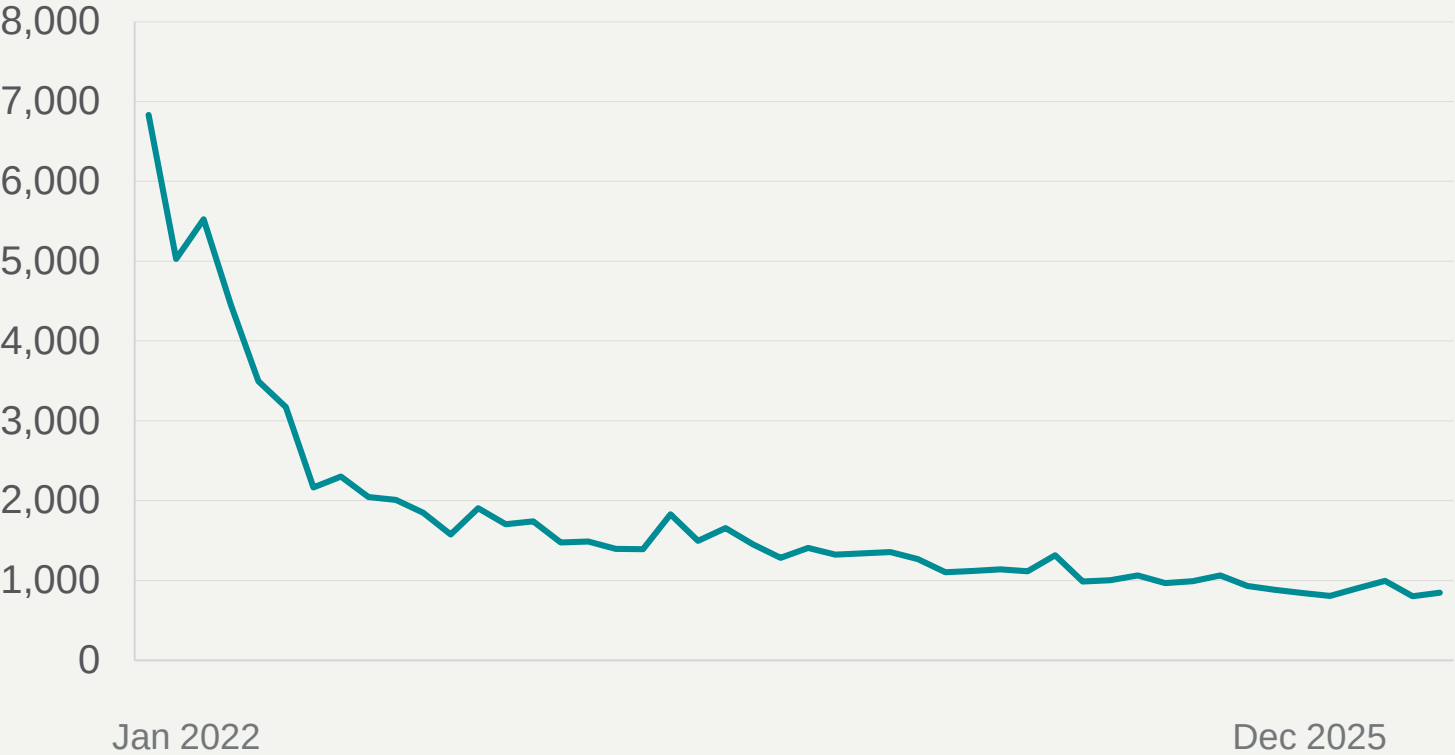
ED first encounters

All have the recorded type
Hospital Encounter.

Full cohort: 84,850 patients. “Other types” combines 11 less frequent categories.

First encounters over time

Monthly volume and annual ED rates

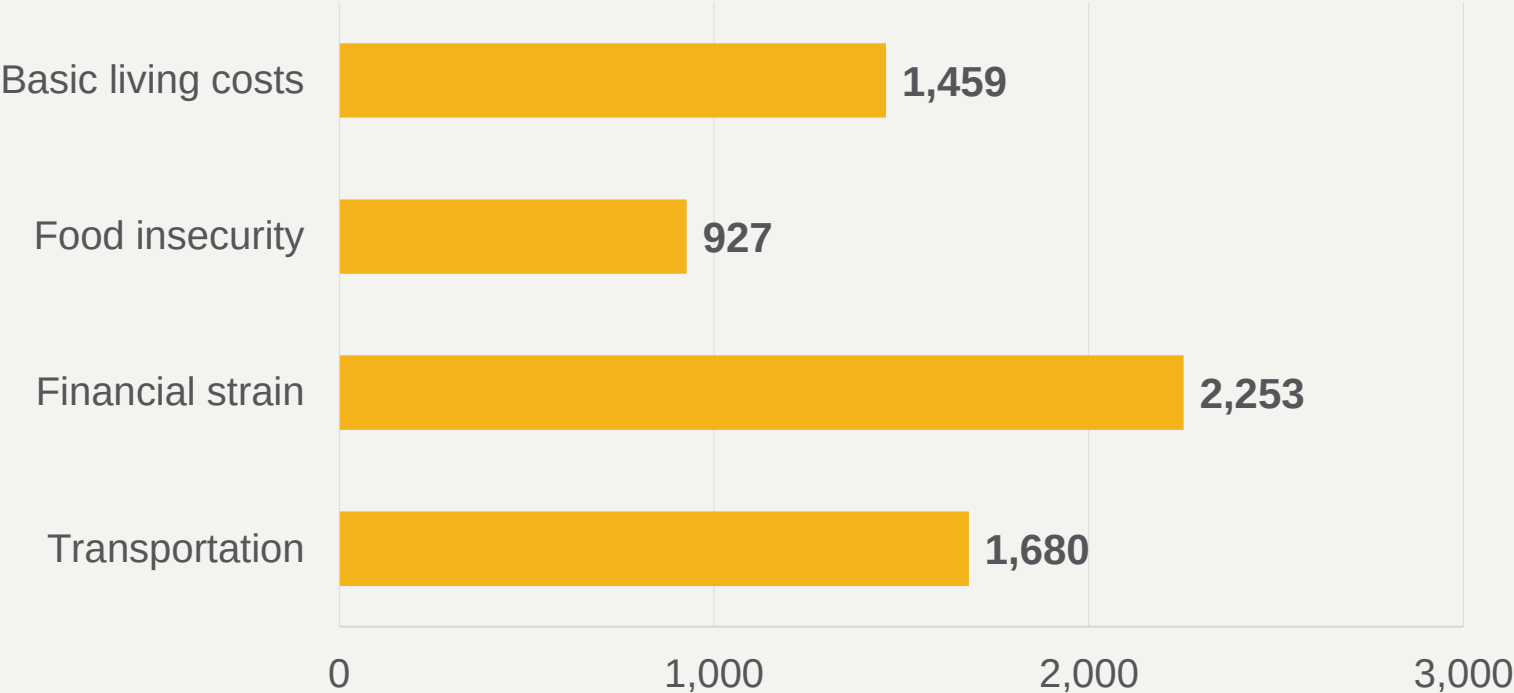


Year	Patients	ED rate
2022	40,453	2.04%
2023	18,822	4.62%
2024	14,476	4.88%
2025	11,099	6.23%

Months run from January 2022 to December 2025. The start of the observation window can affect first-encounter volume.

Recorded social needs

All-period summaries linked to the first-encounter cohort



33,815

patients with linked
all-period summaries

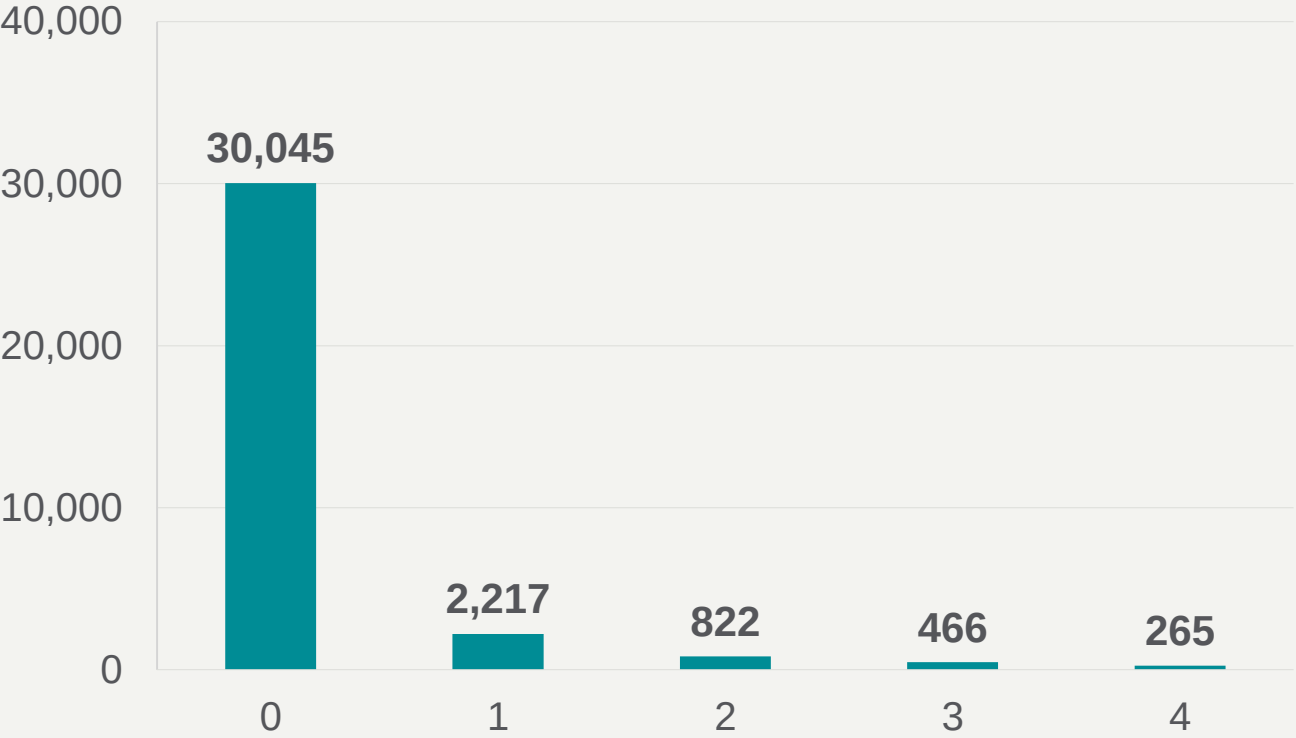
51,035

patients without a
linked summary

Domain counts overlap. No recorded need can include unanswered questions. Records can come from after the first encounter.

Number of recorded need domains

All-period summary patients only

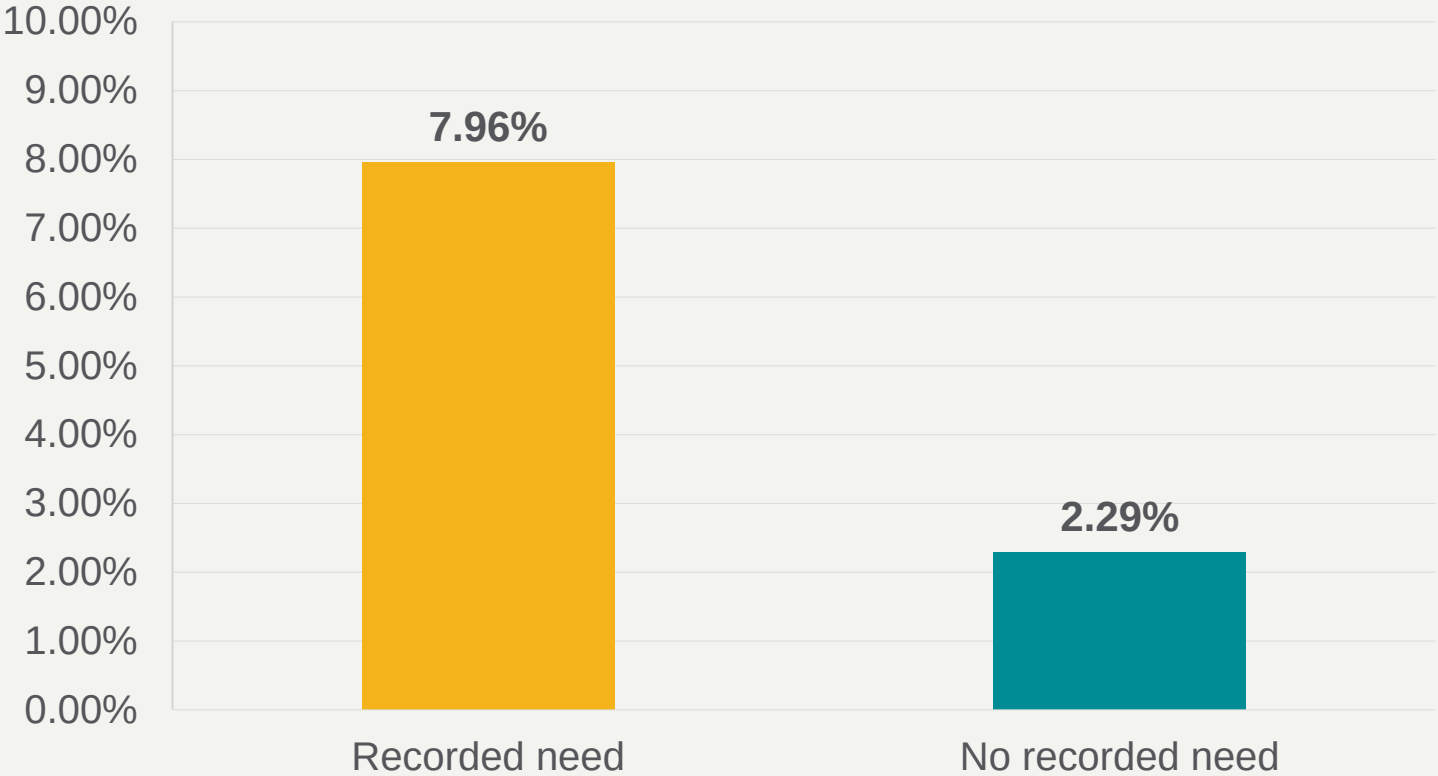


Flagged domains	ED / patients	ED rate
0	688 / 30,045	2.29%
1	118 / 2,217	5.32%
2	72 / 822	8.76%
3	58 / 466	12.45%
4	52 / 265	19.62%

Denominator: 33,815 linked patients. Zero flagged domains does not establish a complete negative screen.

Recorded needs and ED entry

All-period social-need flags

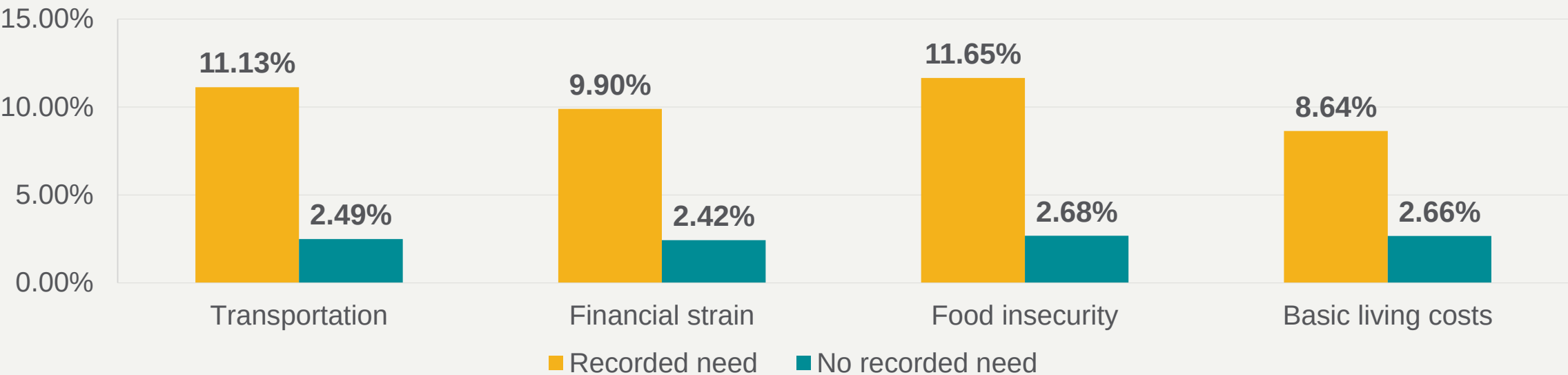


Group	ED / patients
Recorded need	300 / 3,770
No recorded need	688 / 30,045

These groups differ in recorded need status. The comparison does not control for severity, screening selection, or other confounding.

ED rates across social-need domains

All-period flags within the linked cohort

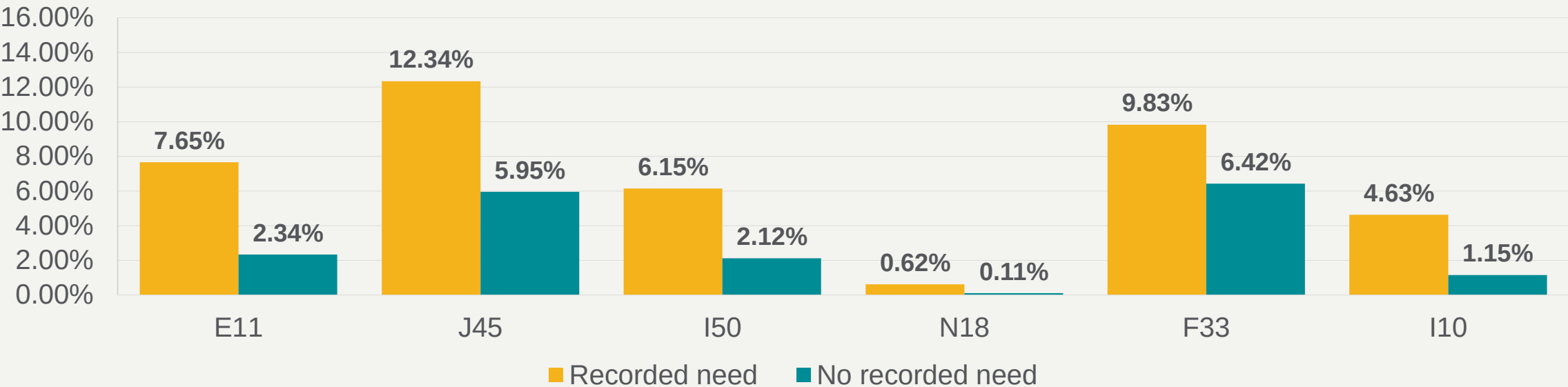


ED / patients	Transportation	Financial strain	Food insecurity	Basic living costs
Recorded need	187 / 1,680	223 / 2,253	108 / 927	126 / 1,459
No recorded need	801 / 32,135	765 / 31,562	880 / 32,888	862 / 32,356

Domains overlap. Each domain comparison uses the same linked cohort, grouped by that domain’s recorded flag.

Disease-specific social-need comparisons

Each rate uses that disease’s first observed encounter and an all-period need flag



ED / patients	E11	J45	I50	N18	F33	I10
Recorded need	105 / 1,372	57 / 462	30 / 488	3 / 487	104 / 1,058	96 / 2,074
No recorded need	230 / 9,843	156 / 2,623	63 / 2,977	5 / 4,731	189 / 2,945	252 / 21,969

E11 diabetes J45 asthma I50 heart failure N18 kidney disease F33 recurrent depression I10 hypertension

ED flags within encounter types

Selected encounter types from the all-period linked cohort

Encounter type	Recorded need ED rate and ED / patients	No recorded need ED rate and ED / patients
Hospital Encounter	36.72% 300 / 817	36.97% 688 / 1,861
Appointment	0.00% 0 / 1	0.00% 0 / 22
Patient Outreach	0.00% 0 / 137	0.00% 0 / 600
Telemedicine	0.00% 0 / 134	0.00% 0 / 636

A 0% ED rate describes these encounters. It does not measure whether outreach or telemedicine prevented later ED visits.

No observations means an empty group. A zero rate means the group has observations but no recorded ED flags.

Coverage before the first encounter

Latest eligible answer per question, strictly before the first-encounter day

1.77%

1,505 of 84,850 patients
have a classifiable need status

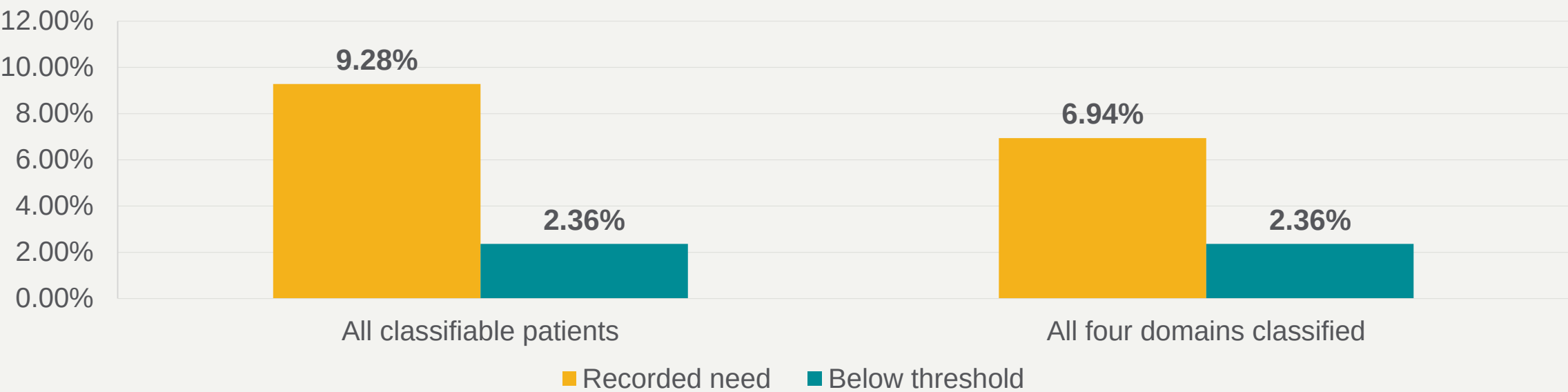
2,410 patients have an eligible response.
Some still lack enough information to classify.

Status	Patients
Recorded need	194
Below threshold	1,311
Unknown	83,345

One positive domain establishes recorded need. Below threshold requires all four domains to be classifiable.

ED rates using prior responses

Primary food threshold: “Often true”



ED / patients	All classifiable	All four domains
Recorded need	18 / 194	10 / 144
Below threshold	31 / 1,311	31 / 1,311

Small, selected groups limit interpretation. Complete-domain restriction changes the positive group and its ED rate.

Food-threshold sensitivity

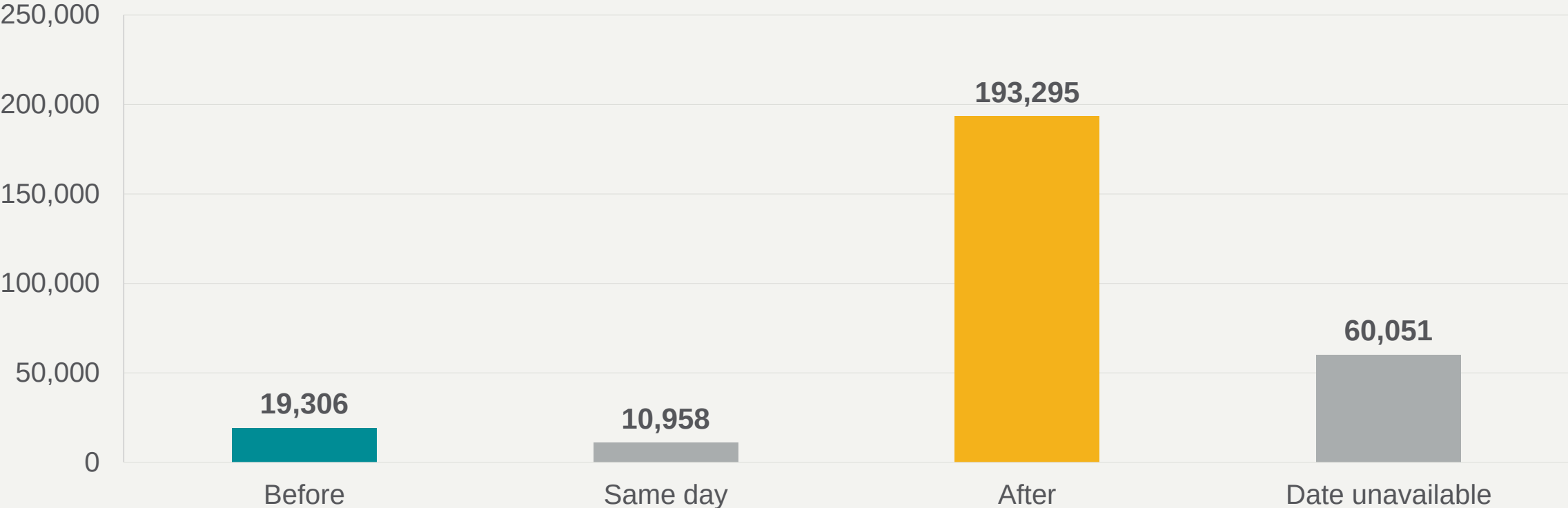
The broader definition adds “Sometimes true” to the food flag

Measure	Primary threshold	Broader threshold
Recorded need	194	233
ED among recorded need	18 / 194 (9.28%)	19 / 233 (8.15%)
Below threshold	1,311	1,290
ED among below threshold	31 / 1,311 (2.36%)	30 / 1,290 (2.33%)
Unknown	83,345	83,327

Both definitions use prior responses. Changing the threshold can change both classification and the comparison denominator.

When the survey responses occurred

Full source cohort, across the four target domains



283,610 response records, not patients. Only the 19,306 earlier records qualify for the prior-response analysis.

What the analysis shows

Findings from first observed encounters and recorded social needs

3.64%

of patients entered through an ED-flagged encounter in the observed chronic-disease cohort.

7.96% vs 2.29%

ED rates with and without recorded need in the all-period linked summaries.

1,505 patients

have classifiable prior-response status. Missing information limits the comparison.

Earlier screening and access support

Proposals to test using current social-need information

Improve screening records

1.77%

1,505 / 84,850 patients have classifiable responses before their first encounter.

Record response dates and missing-answer reasons. Review unanswered items before care.

Track prior-response coverage and completion across all four domains.

Pilot transportation support

11.13% vs 2.49%

ED rates with and without recorded transport need
187 / 1,680 vs 801 / 32,135 patients

Confirm current barriers, then offer transport or appointment support. Consider telemedicine when suitable.

Track support uptake, appointment attendance, and connection to follow-up care.

Asthma follow-up and pilot evaluation

A focused service pilot with clearly defined outcomes

Review asthma care connections

10.81% overall

1,057 / 9,782 asthma patients had an ED first observed encounter, highest of six groups.

12.34% vs 5.95%

With vs without any recorded need
57 / 462 vs 156 / 2,623 patients

Offer follow-up booking support to asthma patients with confirmed current support needs.

Evaluate before scaling

Define a pilot start date, follow-up window, and a suitable comparison group.

Measure appointment attendance and care connections. Evaluate subsequent ED use with differences between groups in mind.

Two or more flagged domains could prompt staff review. This cutoff is not validated for automatic care prioritization.